

Introduction to the Kalman filter

GIZ/OGR workshop at the Rwanda MINECOFIN

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18-26 November, 2024

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Linear projections

The basic setup

- The main purpose of the Kalman filter is to estimate the values of unobserved quantities based on observed quantities
- This can possibly only be done if there's a relationship between the observed and the unobserved quantities
- We will only discuss the case of a linear relationship, i.e. if the observed variable is Y , and the unobserved variables are $X_1, X_2, \dots, X_m$, then we assume that the stochastic relationship is

$$Y = a + h_1X_1 + h_2X_2 + \dots + h_mX_m \quad (1)$$

Non-deterministic relationships

- As the first example, suppose there's only one unobserved variable, and we know the values of the coefficients a and h

$$Y = a + hX \quad (2)$$

- In this case the "estimation" of the unobserved variable is trivial

$$\hat{X} = \frac{Y - a}{h} \quad \text{if } h \neq 0 \quad (3)$$

- For the problem to be non-trivial, this relationship can not be deterministic, i.e. it must involve uncertainty

Linear estimators

- The simplest nontrivial case is when we have two unobserved variables

$$Y = a + h_X X + h_Z Z, \quad h_X \neq 0, h_Z \neq 0 \quad (4)$$

- We are looking for a function f of Y , such that $f(Y)$ is in "some sense" the best estimates of X
- We will further restrict ourself and only deal with linear estimators

$$\hat{X} = f(Y) = \lambda + \kappa Y \quad (5)$$

- The task is now to specify what "in some sense" means, and how the projection parameters λ and κ should be calculated

Linear projections

- A "sense" that certainly makes sense is to look for an estimator which—on average—produces as small estimation errors as possible
- We measure the estimation error by the Root Mean Squared Error (*RMSE*)
- For example, if we have observations of Y_t for $t = 1, \dots, T$, then the *RMSE* of X_t corresponding to λ and κ can be calculated as

$$\begin{aligned} RMSE(\hat{X} \mid \lambda, \kappa) &= \sqrt{\frac{\sum_{t=1}^T (X_t - \hat{X}_t)^2}{T}} = \\ &= \sqrt{\frac{\sum_{t=1}^T (X_t - \lambda - \kappa Y_t)^2}{T}} \end{aligned} \tag{6}$$

Linear projection coefficients

- In other words, λ and κ are the linear regression coefficients when regressing X on Y and a constant
- But we can not possibly run these regressions, since we have no observations for X_t
- So we have to calculate the theoretical regression coefficients, using the linear relationship between Y and X
- The theoretical counterpart of the MSE is

$$MSE(\lambda, \kappa) = E(X - \lambda - \kappa Y)^2 \quad (7)$$

- We want to find theoretical formulas for λ and κ which minimize this sum
- "Theoretical" here means that not in terms of observed Y_t and $X_{1,t}$ data, but their in terms of their means and correlations

Linear projection coefficients

- Let's first expand the formula

$$MSE(\lambda, \kappa) = EX^2 + \lambda^2 + \kappa^2 EY^2 - 2\lambda EX - 2\kappa EXY + 2\lambda\kappa EY$$

- The first order conditions are

$$0 = \lambda - EX + \kappa EY$$

$$0 = \kappa EY^2 - EXY + \lambda EY$$

- Substituting the λ from the first in to the second gives

$$0 = \kappa EY^2 - EXY + (EX - \kappa EY) EY$$

$$0 = \kappa EY^2 - EXY + EXEY - \kappa E^2Y$$

$$\kappa = \frac{EXY - EXEY}{EY^2 - E^2Y} = \frac{\text{Cov}(X, Y)}{\text{Var } Y} \quad (8)$$

- And from here

$$\lambda = EX - \frac{\text{Cov}(X, Y)}{\text{Var } Y} EY \quad (9)$$

The estimation equation as a forecast update

- The estimation of X will therefore be

$$\hat{X} = EX - \frac{\text{Cov}(X, Y)}{\text{Var } Y} EY + \frac{\text{Cov}(X, Y)}{\text{Var } Y} Y$$

- Combining terms we arrive to

$$\hat{X} = EX + \frac{\text{Cov}(X, Y)}{\text{Var } Y} (Y - EY) \quad (10)$$

- This has a very intuitive interpretation in terms of forecast update when new information arrives
 - Before any observation, the best guess of X is EX , and the best guess of Y is EY
 - Then the observation of Y arrives, but the new information is only $Y - EY$
 - $\frac{\text{Cov}(X, Y)}{\text{Var } Y}$ is the update elasticity
 - It depends negatively on the "noisiness" of the observation Y
 - Finally, the old forecast EX is updated by the elasticity $\times$ new information

Calculating the moments

- To use equation (10), we need to know various quantities which appear there
- We can use the "observation equation" (4) to express some expectations in terms of others
- First, let's do EY

$$EY = E(a + h_X X + h_Z Z) = a + h_X EX + h_Z EZ \quad (11)$$

- $\text{Cov}(X, Y)$ is only slightly more involved

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(X, a + h_X X + h_Z Z) = \\ &= h_X \text{Cov}(X, X) + h_Z \text{Cov}(X, Z) = \\ &= h_X \text{Var } X + h_Z \text{Cov}(X, Z) \end{aligned} \quad (12)$$

Calculating moments

- The most complicated is $\text{Var } Y$

$$\begin{aligned}\text{Var } Y &= \text{Var } (a + h_X X + h_Z Z) = \text{Var } a + h_X^2 \text{Var } X + h_Z^2 \text{Var } Z + \\ &\quad + 2 \text{Cov } (a, X) + 2 \text{Cov } (a, Z) + 2h_X h_Z \text{Cov } (X, Z) = \\ &= h_X^2 \text{Var } X + h_Z^2 \text{Var } Z + 2h_X h_Z \text{Cov } (X, Z)\end{aligned}\quad (13)$$

- The estimation (10) can only be used if we assume that we know the following

$$\mu_X = EX$$

$$\mu_Z = EZ$$

$$\sigma_X^2 = \text{Var } X$$

$$\sigma_Z^2 = \text{Var } Z$$

$$\rho = \frac{\text{Cov } (X, Z)}{DX \cdot DZ}$$

Summary

- We observe a random quantity Y
- We know that Y is related to some other random quantities X and Z by $Y = a + h_X X + h_Z Z$
- We want to estimate the values of X using only the observation Y
- We assume that
 - The parameters a , h_X , and h_Z are known
 - The expectations and covariances of X and Z are known

The uncertainty of the estimation

- We can calculate the uncertainty of this estimation as follows

$$\begin{aligned}MSE(\lambda, \kappa) &= E(X - \hat{X})^2 = E(X - \lambda - \kappa Y)^2 = \text{Var}(X - \lambda - \kappa Y) = \\&= \text{Var}(X - \kappa Y) = \text{Var} X + \kappa^2 \text{Var} Y - 2\kappa \text{Cov}(X, Y) = \\&= \text{Var} X + \left(\frac{\text{Cov}(X, Y)}{\text{Var} Y} \right)^2 \text{Var} Y - 2 \frac{\text{Cov}(X, Y)}{\text{Var} Y} \text{Cov}(X, Y)\end{aligned}\tag{14}$$

- We get after simplification that

$$MSE(\lambda, \kappa) = \text{Var} X - \frac{\text{Cov}^2(X, Y)}{\text{Var} Y}\tag{15}$$

- This is again quite intuitive
 - The uncertainty of the updated forecast is always smaller
 - The reduction is smaller when the information is more noisy

Determining the necessary parameters

- How do we get EX , EZ , $\text{Var } X$, $\text{Var } Z$, $\text{Cov}(X, Z)$?
- In practice, these are often determined by the a theoretical model
- Equation (4) becomes the observation/measurement equation of this model
- In most cases, the filtering is applied to time series data
- The theoretical equations will be called the transition equations of the model

Section contents

Time series filtering

Special features of time series: autocorrelation

- The algorithm can be applied to time series data as well
- We will derive formulas only for the simplest case when the observation equation is

$$Y_t = X_t + Z_t \quad (16)$$

with the two unobserved transition variables X_t and Z_t uncorrelated

- With assumptions for EX , EZ , $\text{Var } X$, $\text{Var } Z$, and using the formulas of the previous section, we can derive the estimators for X_t and Z_t as before
- But this estimator will only use the observed value of Y_t to estimate the unobserved variables
- Since time series typically exhibit autocorrelation, observations in all sample periods might (and should) be used

Special features of time series: autocorrelation

- We represent autocorrelation by specifying the transition equation for X_t as

$$X_t = c_X + \phi_X X_{t-1} + \varepsilon_t^X, \quad \text{Var } \varepsilon_t^X = \sigma_X^2 \quad (17)$$

- This readily implies that the mean and variance of X_t are given by

$$EX_t = \frac{c_X}{1 - \phi_X} \quad \text{Var } X_t = \frac{\sigma_X^2}{1 - \phi_X^2} \quad (18)$$

for $0 < \phi_X < 1$

- In this case X_t and Y_s will also be correlated for all $0 < t, s$, and therefore the estimate of X_t should be based on the full sample $Y_1, Y_2, \dots, Y_T$

Special features of time series: autocorrelation

- This means that we should calculate the coefficients in the theoretical regression

$$X_t = \lambda + \sum_{t=1}^T \kappa_t Y_t \quad (19)$$

- And to do this, we need to be able to calculate all possible covariances between X_t and Z_s for $0 < t, s$, besides all auto-covariances of X_t and Z_t
- This in practice may not be so very hard, but to calculate the projection coefficients κ_t from the covariances we have to solve a linear equation of order T , which can be practically prohibiting

Special features of time series: non-stationarity

- Another practically important issue with time series data is non-stationarity
- If for example Z_t is non-stationary, then the unconditional mean and variance of Z_t are not defined
- An equation with a fixed long term growth rate will lead to non-stationary levels
- Working with growth rates implies that out level estimates will be given by accumulating growth estimates from a given initial level
- Therefore the level estimates will depend on the assumed value of the initial level

Special features of time series: recursive estimation

- The solution to both issues (large system, non-stationary) is to work with conditional estimators
- Conditional on some initial values, all necessary means and variances are well defined
- Furthermore, from given initial guesses, we can derive recursive formulas to update the estimations, so instead of solving a huge system of equations, we solve many small systems, which is usually much faster
- We will continue with the example of filtering the trend of GDP

A simple GDP trend/gap model

- Let's consider the following (in fact, the simplest possible) output gap model

$$\tilde{y}_t = \phi \tilde{y}_{t-1} + \tilde{\varepsilon}_t \quad \text{Var } \tilde{\varepsilon}_t = \tilde{\sigma}^2 \quad (20)$$

$$d\bar{y}_t = (1 - \rho) \delta + \rho d\bar{y}_{t-1} + \bar{\varepsilon}_t \quad \text{Var } \bar{\varepsilon}_t = \bar{\sigma}^2 \quad (21)$$

$$d\bar{y}_t = \bar{y}_t - \bar{y}_{t-1} \quad (22)$$

- We have three transition variables: $\tilde{y}_t$, $\bar{y}_t$, and $d\bar{y}_t$
- We have one observation equation

$$y_t = \tilde{y}_t + \bar{y}_t \quad (23)$$

- We assume that the gap and trend shocks are uncorrelated, implying that the gaps and trends are uncorrelated in all periods

The recursive algorithm

- The task is to estimate the gap values from a sample of T observations $y_1, y_2, \dots, y_T$
- We will accomplish this by the following algorithm
 - Assume an initial gap and trend estimates $\tilde{y}_{1|0}, \bar{y}_{1|0}$, and $d\bar{y}_{1|0}$, with uncertainties $\tilde{P}_{1|0}, \bar{P}_{1|0}$, and $d\bar{P}_{1|0}$
 - During a first iteration, which runs forward from $t = 1$ to $t = T$, we will get real-time estimates of $\tilde{y}_t, \bar{y}_t$, and $d\bar{y}_t$
 - During the second iteration, which runs backwards from $t = T$ to $t = 1$, we will get are final gap and trend estimates
- But what does "real time estimate" exactly means

Filtering and smoothing

- We are ultimately interested in the gap/trend estimates based on the information of the full sample; these so called *smoothed* will be denoted by

$$\tilde{y}_{t|T} \quad \bar{y}_{t|T} \quad d\bar{y}_{t|T} \quad (24)$$

- During the calculations, however, we will also need estimates which are based on a partial information set; in particular we will need to calculate "real time" estimates for period- t values base on observations only up to period t ; these are called *filtered* estimates, and are denoted by

$$\tilde{y}_{t|t} \quad \bar{y}_{t|t} \quad d\bar{y}_{t|t} \quad (25)$$

- Finally, in some formulas we will also need *predicted* estimates, denoted by

$$\tilde{y}_{t|t-1} \quad \bar{y}_{t|t-1} \quad d\bar{y}_{t|t-1} \quad (26)$$

Updating linear projections

- We will use the same two equations (10) and (15) as before

$$\hat{X} = EX + \frac{\text{Cov}(X, Y)}{\text{Var } Y} (Y - EY) \quad (27)$$

$$MSE = \text{Var } X - \frac{\text{Cov}^2(X, Y)}{\text{Var } Y} \quad (28)$$

- Remember the intuitive content
 - EX is an old estimate of X and $\text{Var } X$ is its uncertainty (variance)
 - The observation is Y and the new information is $Y - EY$; its variance is $\text{Var } Y$
 - The two equations give the new projection updated the new information, and the uncertainty of the updated projection

Updating equations with projection errors

- Using the definition of the variance, and the fact that correlations are not affected by adding/subtracting constant, we can write

$$\hat{X} = EX + \frac{\text{Cov}(X - EX, Y - EY)}{E(Y - EY)^2} (Y - EY) \quad (29)$$

$$MSE = E(X - EX)^2 - \frac{\text{Cov}^2(X - EX, Y - EX)}{E(Y - EY)^2} \quad (30)$$

- The purpose of this is that the updating equations are now expressed solely in terms of the projection errors $Y - EY$ and $X - EX$
- This is the form which can be generalized into a dynamic recursive setting

The matrix form of the model

- Unfortunately, even in this very simple case, we can not avoid working with the matrix form of the model
- For this, we need to rearrange the last equations a bit

$$\begin{aligned}d\bar{y}_t &= \bar{y}_t - \bar{y}_{t-1} \\ \bar{y}_t &= \bar{y}_{t-1} + d\bar{y}_t \\ \bar{y}_t &= \bar{y}_{t-1} + (1 - \rho) \delta + \rho d\bar{y}_{t-1} + \bar{\varepsilon}_t\end{aligned}\tag{31}$$

- Then the matrix form will be

$$\begin{pmatrix} \tilde{y}_t \\ d\bar{y}_t \\ \bar{y}_t \end{pmatrix} = \begin{pmatrix} 0 \\ (1 - \rho) \delta \\ (1 - \rho) \delta \end{pmatrix} + \begin{pmatrix} \phi & 0 & 0 \\ 0 & \rho & 0 \\ 0 & \rho & 1 \end{pmatrix} \begin{pmatrix} \tilde{y}_{t-1} \\ d\bar{y}_{t-1} \\ \bar{y}_{t-1} \end{pmatrix} + \begin{pmatrix} \tilde{\varepsilon}_t \\ \bar{\varepsilon}_t \\ \bar{\varepsilon}_t \end{pmatrix}\tag{32}$$

- And the measurement equation will be

$$y_t = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{y}_t \\ d\bar{y}_t \\ \bar{y}_t \end{pmatrix}\tag{33}$$

The matrix from of the model

- Once again we simplify notation and write

$$\xi_t = c + F\xi_{t-1} + \eta_t \quad \text{Var}(\eta_t) = Q \quad (34)$$

$$y_t = H\xi_t \quad (35)$$

where

$$\xi_t = \begin{pmatrix} \tilde{y}_t \\ d\bar{y}_t \\ \bar{y}_t \end{pmatrix} \quad \eta_t = \begin{pmatrix} \tilde{\varepsilon}_t \\ \bar{\varepsilon}_t \\ \bar{\varepsilon}_t \end{pmatrix} \quad (36)$$

and

$$c = \begin{pmatrix} 0 \\ (1-\rho)\delta \\ (1-\rho)\delta \end{pmatrix}, \quad F = \begin{pmatrix} \phi & 0 & 0 \\ 0 & \rho & 0 \\ 0 & \rho & 1 \end{pmatrix}, \quad H = (1 \quad 0 \quad 1), \quad Q = \begin{pmatrix} \tilde{\sigma}^2 & 0 & 0 \\ 0 & \bar{\sigma}^2 & \bar{\sigma}^2 \\ 0 & \bar{\sigma}^2 & \bar{\sigma}^2 \end{pmatrix} \quad (37)$$

The state space form

- In practical (mostly engineering) application, it is also common to add explicit measurement errors, so that the model becomes

$$\xi_t = c + F\xi_{t-1} + \eta_t \quad \text{Var}(\eta_t) = Q \quad (38)$$

$$y_t = H\xi_t + \zeta_t \quad \text{Var}(\zeta_t) = R \quad (39)$$

- We assume that all measurement errors are uncorrelated with any other shock/variable in the model
- In what follows we will derive the Kalman filter equations for this more general setting, even though we have no measurement errors

Updating equations in matrix form

- We also have to make sure that the updating equations are valid in the multivariate setting
- But after an easy modification everything will work for matrices
- We simplify the notation to avoid overcrowded formulas:

$$C_{YX} = \text{Cov}(X - EX, Y - EY)$$

$$V_X = \text{Var}(X - EX)$$

$$V_Y = \text{Var}(Y - EY)$$

- With this, the matrix form will be

$$\hat{X} = EX + C_{YX}V_Y^{-1}(Y - EY) \quad (40)$$

$$MSE = V_X - C_{YX}V_Y^{-1}C'_{YX} \quad (41)$$

The Kalman filter: forward iterations

- We start with initial guess $\xi_{1|0}$, and an associated variance $P_{1|0}$
- Suppose the forward iterations have already given us $\xi_{t|t-1}$ and $P_{t|t-1}$ with

$$P_{t|t-1} = \text{Var}(\xi_t - \xi_{t|t-1}) \quad (42)$$

- These are the old forecast and its uncertainty
- The new information in period t is $y_t - y_{t|t-1}$
- To apply the update equation we need to express

$$y_{t|t-1} \quad \text{Cov}(\xi_t - \xi_{t|t-1}, \quad y_t - y_{t|t-1}) \quad \text{Var}(y_t - y_{t|t-1})$$

using the model parameters and the current estimates

The Kalman filter: forward iterations: update step

- The first is easy

$$y_{t|t-1} = H\xi_{t|t-1} \quad (43)$$

- Using the measurement equation we get that

$$\begin{aligned} & \text{Cov}(\xi_t - \xi_{t|t-1}, y_t - y_{t|t-1}) \\ &= \text{Cov}(\xi_t - \xi_{t|t-1}, H\xi_t + \zeta_t - H\xi_{t|t-1}) \\ &= \text{Cov}(\xi_t - \xi_{t|t-1}, H(\xi_t - \xi_{t|t-1})) \\ &= \text{Cov}(\xi_t - \xi_{t|t-1}, \xi_t - \xi_{t|t-1}) H' \\ &= P_{t|t-1} H' \end{aligned} \quad (44)$$

The Kalman filter: forward iterations: update step

- Similarly

$$\begin{aligned}
 & \text{Cov}(y_t - y_{t|t-1}, \quad y_t - y_{t|t-1}) \\
 &= \text{Cov}(H\xi_t + \zeta_t - H\xi_{t|t-1}, \quad H\xi_t + \zeta_t - H\xi_{t|t-1}) \\
 &= \text{Cov}(H(\xi_t - \xi_{t|t-1}), \quad H(\xi_t - \xi_{t|t-1})) + \text{Var} \zeta_t \\
 &= H \text{Cov}(\xi_t - \xi_{t|t-1}, \quad \xi_t - \xi_{t|t-1}) H' + \text{Var} \zeta_t \\
 &= HP_{t|t-1} H' + R
 \end{aligned} \tag{45}$$

- The update equations are therefore

$$\xi_{t|t} = \xi_{t|t-1} + P_{t|t-1} H' (HP_{t|t-1} H' + R)^{-1} (y_t - H\xi_{t|t-1}) \tag{46}$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} H' (HP_{t|t-1} H' + R)^{-1} HP_{t|t-1} \tag{47}$$

The Kalman filter: forward iterations: forecast step

- Now, to continue with the forward iterations, we need to calculate $\xi_{t+1|t}$ and $P_{t+1|t}$
- The forecast is easily calculated from the transition equations

$$\xi_{t+1|t} = c + F\xi_{t|t} + \eta_{t+1|t} = c + F\xi_{t|t} \quad (48)$$

- And its variance is only a bit more complicated

$$\begin{aligned} P_{t+1|t} &= \text{Var}(\xi_{t+1} - \xi_{t+1|t}) = \text{Var}(c + F\xi_t + \eta_{t+1} - (c + F\xi_{t|t})) \\ &= \text{Var}(F\xi_t + \eta_{t+1} - F\xi_{t|t}) = \text{Var}(F(\xi_t - \xi_{t|t}) + \eta_{t+1}) \\ &= \text{Var}(F(\xi_t - \xi_{t|t})) + \text{Var}(\eta_{t+1}) \\ &= FP_{t|t}F' + Q \end{aligned} \quad (49)$$

The Kalman filter: forward iterations

- The forward iterations are therefore given by the following four equations

$$\xi_{t|t} = \xi_{t|t-1} + P_{t|t-1}H' (HP_{t|t-1}H' + R)^{-1} (y_t - H\xi_{t|t-1}) \quad (50)$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1}H' (HP_{t|t-1}H' + R)^{-1} HP_{t|t-1} \quad (51)$$

$$\xi_{t+1|t} = c + F\xi_{t|t} \quad (52)$$

$$P_{t+1|t} = FP_{t|t}F' + Q \quad (53)$$

- We can now complete the forward iterations, and calculate real-time gap and trend estimates $\tilde{y}_{t|t}$ and $\bar{y}_{t|t}$

The Kalman filter: backward iterations

- But we want the smoothed, i.e. full information based estimates $\tilde{y}_{t|T}$ and $\bar{y}_{t|T}$
- Well, we have them in the last period T , together with variance matrix $P_{T|T}$
- For $t < T$, we calculate the smoothed estimates in the smoothing steps of the backward iterations
- In a smoothing step we go from $\xi_{t+1|T}$ to $\xi_{t|T}$, and similarly for the variance

The Kalman filter: backward iterations

- We first derive the equation assuming that ξ_{t+1} is exactly known, and then substitute $\xi_{t+1|T}$ for ξ_{t+1}
- The main idea is the same: we want to update the old estimate $\xi_{t|t}$ using the "new observation" ξ_{t+1}
- The term "new" here means relative to the information up to period t , so the new information is $\xi_{t+1} - \xi_{t+1|t}$
- To apply the general update equation we need to calculate

$$\text{Cov}(\xi_t - \xi_{t|t}, \xi_{t+1} - \xi_{t+1|t}) \quad \text{and} \quad \text{Var}(\xi_{t+1} - \xi_{t+1|t}) \quad (54)$$

- The second is simply $P_{t+1|t}$, while the first is

$$\begin{aligned} & \text{Cov}(\xi_t - \xi_{t|t}, \xi_{t+1} - \xi_{t+1|t}) \\ &= \text{Cov}(\xi_t - \xi_{t|t}, c + F\xi_t + \eta_{t+1} - c - F\xi_{t|t}) \\ &= \text{Cov}(\xi_t - \xi_{t|t}, F\xi_t - F\xi_{t|t}) = P_{t|t}F' \end{aligned} \quad (55)$$

The Kalman filter: backward iterations

- We get that

$$\xi_{t|t, \xi_{t+1}} = \xi_{t|t} + P_{t|t} F' P_{t+1|t}^{-1} (\xi_{t+1} - \xi_{t+1|t}) \quad (56)$$

- The final equation of the smoothing step is obtained by replacing ξ_{t+1} with its estimate $\xi_{t+1|T}$

$$\xi_{t|T} = \xi_{t|t} + P_{t|t} F' P_{t+1|t}^{-1} (\xi_{t+1|T} - \xi_{t+1|t}) \quad (57)$$

- The associated variance $P_{t|T}$ could also be derived, but we won't need it now

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The Hodrick-Prescott filter

A very special case

- Consider the model

$$\tilde{y}_t = \tilde{\varepsilon}_t \quad (58)$$

$$d\bar{y}_t = (1 - \rho) \delta + \rho d\bar{y}_{t-1} + \bar{\varepsilon}_t \quad (59)$$

$$d\bar{y}_t = \bar{y}_t - \bar{y}_{t-1} \quad (60)$$

and

$$y_t = \bar{y}_t + \tilde{y}_t \quad (61)$$

- We have seen that the state space coefficients are

$$c = \begin{pmatrix} 0 \\ (1 - \rho) \delta \\ (1 - \rho) \delta \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & \rho & 1 \end{pmatrix}, \quad H = (1 \quad 0 \quad 1), \quad Q = \begin{pmatrix} \tilde{\sigma}^2 & 0 & 0 \\ 0 & \bar{\sigma}^2 & \bar{\sigma}^2 \\ 0 & \bar{\sigma}^2 & \bar{\sigma}^2 \end{pmatrix} \quad (62)$$

together with

$$R = 0 \quad (63)$$

Another state space from

- The measurement equation is

$$y_t = \bar{y}_t + \tilde{y}_t = \bar{y}_t + \tilde{\varepsilon}_t \quad (64)$$

- Since $\tilde{\varepsilon}_t$ is uncorrelated with all other shocks/variables in the model, it can also be interpreted as a measurement error
- This is true in general: the state space form is not unique
- With this interpretation the model becomes

$$d\bar{y}_t = (1 - \rho) \delta + \rho d\bar{y}_{t-1} + \bar{\varepsilon}_t \quad (65)$$

$$\bar{y}_t = \bar{y}_{t-1} + (1 - \rho) \delta + \rho d\bar{y}_{t-1} + \bar{\varepsilon}_t \quad (66)$$

and

$$y_t = \bar{y}_t + \tilde{\varepsilon}_t \quad (67)$$

- Note that this reformulation would not be possible if $\tilde{y}_t$ were autocorrelated

Another state space form

- In this formulation the state space coefficients will be

$$c = \begin{pmatrix} (1 - \rho) \delta \\ (1 - \rho) \delta \end{pmatrix}, \quad F = \begin{pmatrix} \rho & 0 \\ \rho & 1 \end{pmatrix}, \quad H = (0, 1), \quad Q = \begin{pmatrix} \overline{\sigma}^2 & \overline{\sigma}^2 \\ \overline{\sigma}^2 & \overline{\sigma}^2 \end{pmatrix} \quad (68)$$

together with

$$R = \tilde{\sigma}^2 \quad (69)$$

- Running the smoother with these parameters will produce exactly the same result

Direct derivation of the smoothed estimate

- If we set $\rho = 1$, then the Kalman smoother will deliver the same result as the Hodrick - Prescott filter
- However, the easiest way to see this is to return to the full sample setup (10)
- Also, in this case it's more comfortable to work with a single AR(2) state equation

$$\bar{y}_t = c + \phi_1 \bar{y}_{t-1} + \phi_2 \bar{y}_{t-2} + \bar{\varepsilon}_t \quad (70)$$

$$y_t = \bar{y}_t + \tilde{\varepsilon}_t \quad (71)$$

- The second equation is the observation equation, and the output gap is just a measurement shock

Direct derivation of the smoothed estimate

- In the full sample, or "stacked time" setting, we have to calculate the $T \times T$ covariance matrix of

$$X = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_T \end{pmatrix} \in \mathbb{R}^T \quad Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_T \end{pmatrix} \in \mathbb{R}^T \quad (72)$$

together with the variance matrix of Y (also $T \times T$)

- The (t, s) element of $\text{Cov}(X, Y)$ is

$$\text{Cov}(X_t, Y_s) = \text{Cov}(X_t, X_s + \tilde{\varepsilon}_s) = \text{Cov}(X_t, X_s) \quad (73)$$

- And the (t, s) element of $\text{Var}(Y)$ is

$$\text{Cov}(Y_t, Y_s) = \text{Cov}(X_t + \tilde{\varepsilon}_t, X_s + \tilde{\varepsilon}_s) = \begin{cases} \text{Cov}(X_t, X_s) & \text{if } t \neq s \\ \text{Cov}(X_t, X_t) + \tilde{\sigma}^2 & \text{if } t = s \end{cases} \quad (74)$$

Direct derivation of the smoothed estimate

- In other words

$$\text{Var}(Y) = \text{Var}(X) + \tilde{\sigma}^2 I_T \quad (75)$$

where I_T is the $T \times T$ identity matrix

- The projection matrix is then

$$\begin{aligned} P &= \text{Cov}(X, Y) \text{Var}^{-1}(Y) = \text{Var}(X) (\text{Var}(X) + \tilde{\sigma}^2 I_T)^{-1} \\ &= (I_T + \tilde{\sigma}^2 \text{Var}^{-1}(X))^{-1} \end{aligned} \quad (76)$$

Direct derivation of the smoothed estimate

- However, it is not straightforward to calculate these covariances, let alone the inverse of $\text{Var}(Y)$
- So we use a trick; let's define the $T \times T$ matrix L as

$$L = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ -\phi_1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ -\phi_2 & -\phi_1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_2 & -\phi_1 & 1 \end{pmatrix} \quad (77)$$

Direct derivation of the smoothed estimate

- Let's next calculate LX

$$\begin{aligned}
 LX &= \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ -\phi_1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ -\phi_2 & -\phi_1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_2 & -\phi_1 & 1 \end{pmatrix} \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \bar{y}_3 \\ \vdots \\ \bar{y}_{T-2} \\ \bar{y}_{T-1} \\ \bar{y}_T \end{pmatrix} \\
 &= \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 - \phi_1 \bar{y}_1 \\ \bar{y}_3 - \phi_1 \bar{y}_1 - \phi_2 \bar{y}_1 \\ \vdots \\ \bar{y}_{T-2} - \phi_1 \bar{y}_{T-3} - \phi_2 \bar{y}_{T-4} \\ \bar{y}_{T-1} - \phi_1 \bar{y}_{T-2} - \phi_2 \bar{y}_{T-3} \\ \bar{y}_T - \phi_1 \bar{y}_{T-1} - \phi_2 \bar{y}_{T-2} \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 - \phi_1 \bar{y}_1 \\ \bar{\varepsilon}_3 \\ \vdots \\ \bar{\varepsilon}_{T-2} \\ \bar{\varepsilon}_{T-1} \\ \bar{\varepsilon}_T \end{pmatrix}
 \end{aligned} \tag{78}$$

Direct derivation of the smoothed estimate

- The covariance matrix of LX is much easier to calculate

$$\text{Cov}(LX, LX) = \begin{pmatrix} S & 0 & 0 & \cdots & 0 & 0 \\ 0 & \bar{\sigma}^2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \bar{\sigma}^2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \bar{\sigma}^2 & 0 \\ 0 & 0 & 0 & \cdots & 0 & \bar{\sigma}^2 \end{pmatrix} = D \quad (79)$$

with a 2×2 block S in the top left corner

- On the other hand

$$D = \text{Cov}(LX, LX) = L \text{Cov}(X, X) L' \quad (80)$$

and therefore the covariance matrix of X and its inverse can be calculated as

$$\text{Var}(X) = L^{-1} D (L')^{-1}, \quad \text{Var}^{-1}(X) = L' D^{-1} L \quad (81)$$

Direct derivation of the smoothed estimate

- Now clearly

$$\begin{aligned}
 D^{-1} &= \begin{pmatrix} S^{-1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & \frac{1}{\sigma^2} & 0 & \cdots & 0 & 0 \\ 0 & 0 & \frac{1}{\sigma^2} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{\sigma^2} & 0 \\ 0 & 0 & 0 & \cdots & 0 & \frac{1}{\sigma^2} \end{pmatrix} \\
 &= \frac{1}{\sigma^2} \begin{pmatrix} \left(\frac{S}{\sigma^2}\right)^{-1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix}
 \end{aligned} \tag{82}$$

Direct derivation of the smoothed estimate

- The S matrix in the top left corner is the variance of the initial trend values
- Without observing the data, the model does not tell anything about the trend level
- So it seems appropriate to have a very low precision/high variance of the initial level estimate
- That means it makes sense to assume that $S^{-1} = 0$
- In which case the first two row/columns of D^{-1} will be all 0

Direct derivation of the smoothed estimate

- Hence we can ignore the first two rows of L in the product
- The final form of the projection matrix will therefore be

$$P = \left(I_T + \frac{\tilde{\sigma}^2}{\sigma^2} L' L \right)^{-1} \quad (83)$$

with

$$L = \begin{pmatrix} -\phi_2 & -\phi_1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & -\phi_2 & -\phi_1 & 1 \end{pmatrix} \quad (84)$$

so that L is $T - 2 \times T$

The conventional HP filter

- The Hodrick-Prescott filter calculates the trend $\bar{y}_t$ of a times series y_t by minimizing the following sum

$$\sum_{t=1}^T (y_t - \bar{y}_t)^2 + \lambda \sum_{t=2}^T (\bar{y}_t - 2\bar{y}_{t-1} + \bar{y}_{t-2})^2 \quad (85)$$

- The second term is the squared sum of the change in growth rates

$$\bar{y}_t - 2\bar{y}_{t-1} + \bar{y}_{t-2} = (\bar{y}_t - \bar{y}_{t-1}) - (\bar{y}_{t-1} - \bar{y}_{t-2}) \quad (86)$$

- So the HP filter is a trade-off between fit and smoothness of the trend, and the parameter λ is the relative weight of the latter

The conventional HP filter

- We can write the objective function using the earlier notation as

$$\begin{aligned} & (Y - X)' (Y - X) + \lambda (LX)' (LX) = \\ & = Y'Y - Y'X - X'Y + X' (I_T + \lambda L'L) X \end{aligned} \tag{87}$$

- The first order condition with respect to X is

$$-2Y' + 2X' (I_T + \lambda L'L) = 0 \tag{88}$$

- Which gives the HP trend as

$$X = (I_T + \lambda L'L)^{-1} Y \tag{89}$$

- We get the same as before, with λ taking on the role of the relative standard deviations